Reproduction of Zhang, Li, Hu & Luo (IEEE T-ED 2009)

"Field Enhancement for Dielectric Layer of HV Devices on SOI"
IEEE Trans. Electron Devices 56(10), pp. 2327-2334 -- the ENDIF principle

IDEA (ENDIF)
  Vertical breakdown is limited by the SILICON critical field E_S,C,
  but the buried DIELECTRIC holds a much larger field E_I set by
  displacement continuity: eps_I E_I = eps_S E_S,C + q sigma_in.
  Three knobs raise E_I (and V_B,V): thin Si (higher E_S,C), low-k
  buried dielectric, and interface charge sigma_in.

KEY RESULTS (model vs paper -- all match to rounding)

Quantity	Model	Paper	
E_S,C at t_S=0.1um	140.8	141	V/um
E_I (t_S=1, eps=3.9, sig=0)	161	161	V/um
E_I (t_S=1, eps=2.0, low-k)	313	319	V/um
E_I at t_S=0.1um	422	423	V/um
V_B,V at t_S=1um	348	348	V
V_B,V at t_S=0.1um	852	852	V
V_B,V (t_S=1, sig=4e12)	719	719	V

SCOPE
  The paper's analytical core (Figs 2-5). Figs 6-8 are specific 2-D
  device structures (step-thickness / VK-LK / trench SOI) vs MEDICI +
  experiment -- out of scope for this analytical reproduction.

Theory -- the ENDIF model

1. Displacement continuity (why E_I is enhanced)

Normal displacement is continuous up to an interface sheet charge:

$$\varepsilon_I E_I = \varepsilon_S E_{S,C} + q\sigma_{in} \Rightarrow E_I = \frac{\varepsilon_S}{\varepsilon_I} E_{S,C} + \frac{q\sigma_{in}}{\varepsilon_I}$$

Since eps_S/eps_I ~ 11.7/3.9 ~ 3, the dielectric already carries ~3x the silicon field; low-k (smaller eps_I) or interface charge raise it.

2. Silicon critical field vs thickness (eq.10, t_S in um)

$$E_{S,C}(t_S) = 9.783 \frac{21.765 - \ln t_S}{\ln t_S + 3.975} \times 10^4 \text{ V/cm}$$

rises sharply for thin silicon -- 141 V/um at t_S=0.1um, far above the conventional constant 20 V/um. alpha_S,C = A exp(-B/E_S,C) is the Chynoweth rate at E_S,C (A=7.03e5/cm, B=1.231e6 V/cm).

3. Vertical breakdown voltage (eq.13)

$$V_{B,V} = \left\{ \left[9.783 \frac{21.765 - \ln t_S}{\ln t_S + 3.975} \right] (\varepsilon_S t_I + \tfrac{1}{2}\varepsilon_I t_S) + q\sigma_{in} t_I \times 10^{-4} \right\} / \varepsilon_I$$

U-shaped in t_S: high for thin silicon (large E_S,C) and again for thick silicon; larger sigma_in and smaller eps_I lift the whole family.

4. Enhancement factors (eq.12, E_I,Con=60 V/um)

$$\Delta\gamma_1 = \frac{\varepsilon_S \Delta E_{S,C}}{\varepsilon_{SiO_2} E_{I,Con}}, \quad \Delta\gamma_2 = \frac{\Delta\varepsilon_I}{\varepsilon_{SiO_2} - \Delta\varepsilon_I}, \quad \Delta\gamma_3 = \frac{q\sigma_{in}}{\varepsilon_{SiO_2} E_{I,Con}}$$

each reaches 15.67 (E_I=1000 V/um) alone at t_S<50nm, eps_I=0.25eps0, or sigma_in=2e13 -- so any ONE knob suffices; interface charge is the most practical.

Fig. 2 -- E_S,C and alpha_S,C vs t_S

Paper (original)

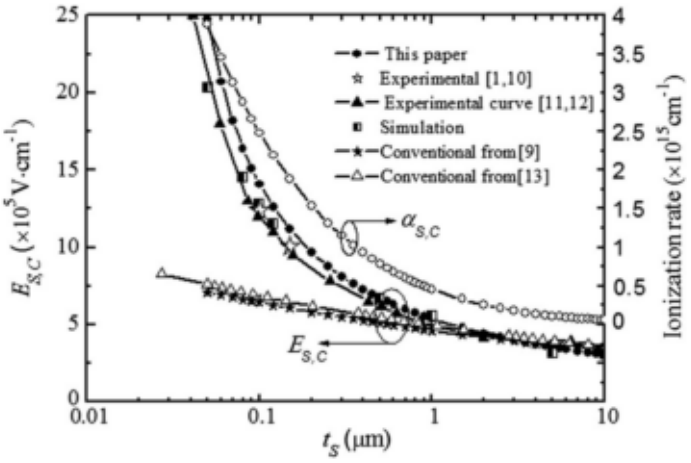
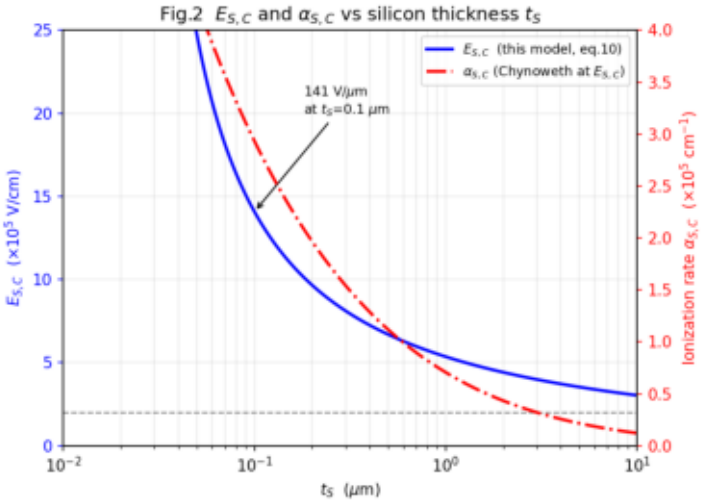


Fig. 2. Dependences of $E_{S,C}$ and $\alpha_{S,C}$ on t_S. Several analytical, simulative, and experimental results are compared. Results from [11]–[13] are obtained by

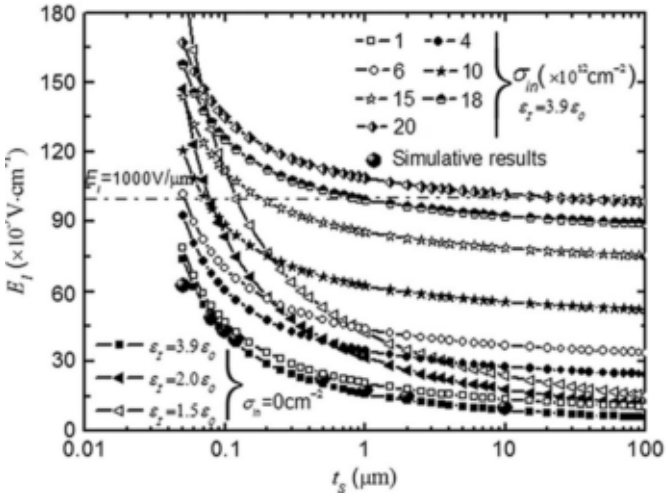
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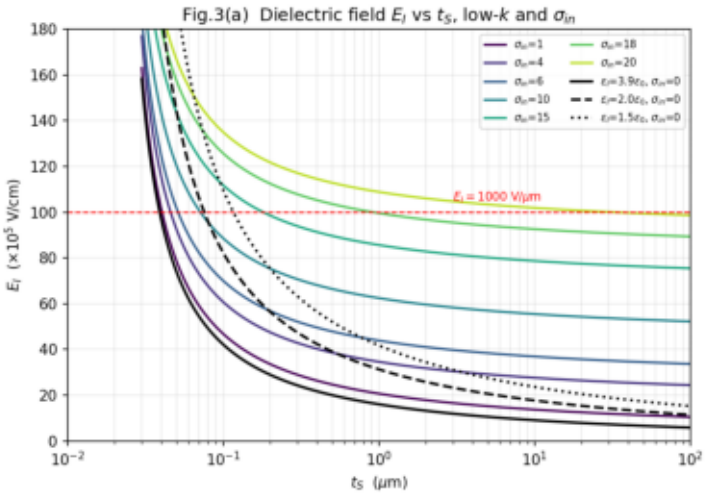
Critical field rises steeply for thin silicon (141 V/um at 0.1um); the critical ionisation rate alpha_S,C (right axis, Chynoweth at E_S,C) grows as t_S shrinks.

Fig. 3(a) -- dielectric field E_I vs t_S, low-k and sigma_in

Paper (original)



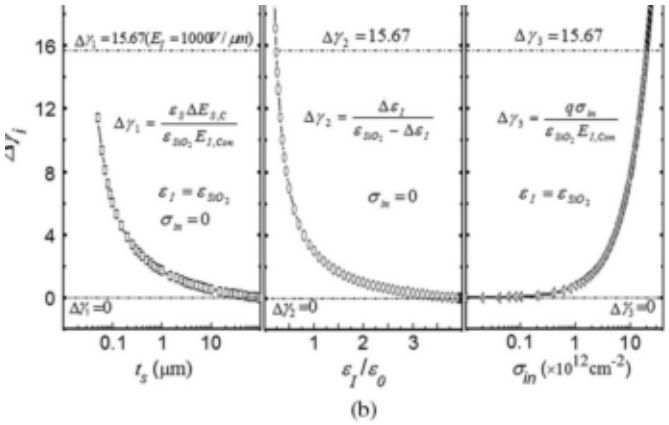
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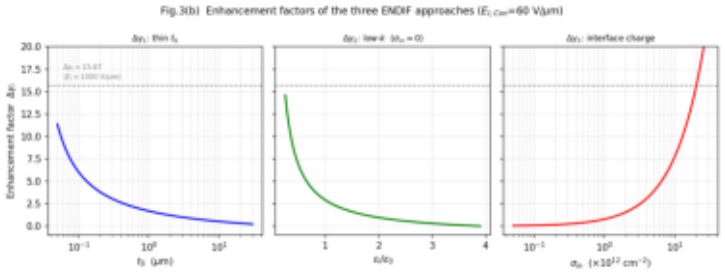
Upper family: sigma_in=1..20e12 (eps_I=3.9eps0). Lower black curves: eps_I=3.9/2.0/1.5 eps0 (sigma_in=0). Both knobs push E_I toward the 1000 V/um dashed line.

Fig. 3(b) -- enhancement factors

Paper (original)



This reproduction



Each approach reaches delta-gamma_i=15.67 (E_I=1000 V/um) on its own: thin t_S, low eps_I, or interface charge (crossing at ~2e13, matching the paper).

Fig. 4 -- vertical breakdown voltage V_B,V vs t_S

Paper (original)

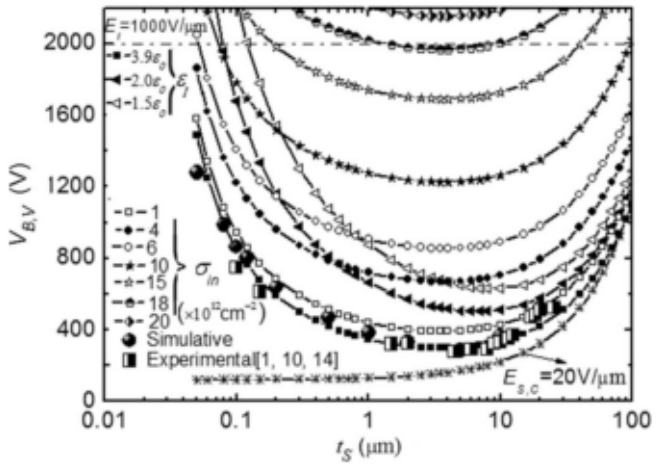
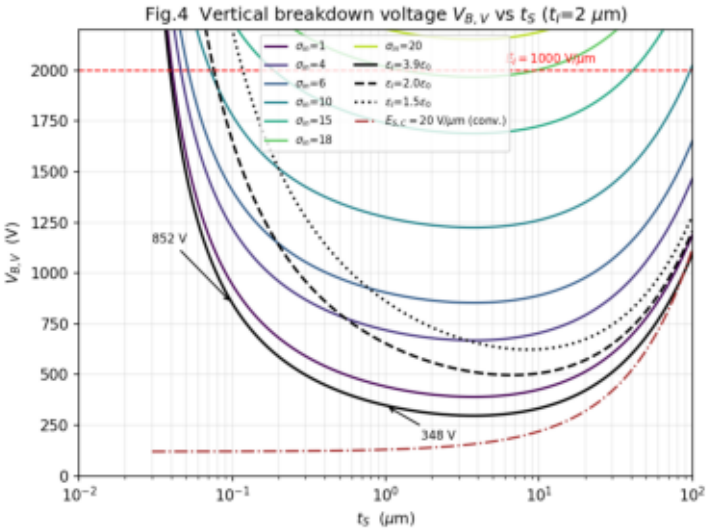


Fig. 4. Dependence of $V_{B,V}$ on t_S, low ε_I, and σ_{in} in the analytical, simulative, and experimental results. Linear impurity in the silicon layer is used while simulating, $N_{dn} = (1.5n + 1)N_{d0}$, where n is the number of

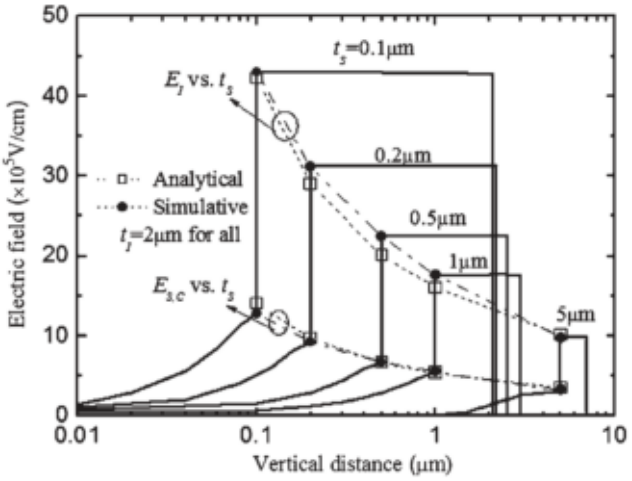
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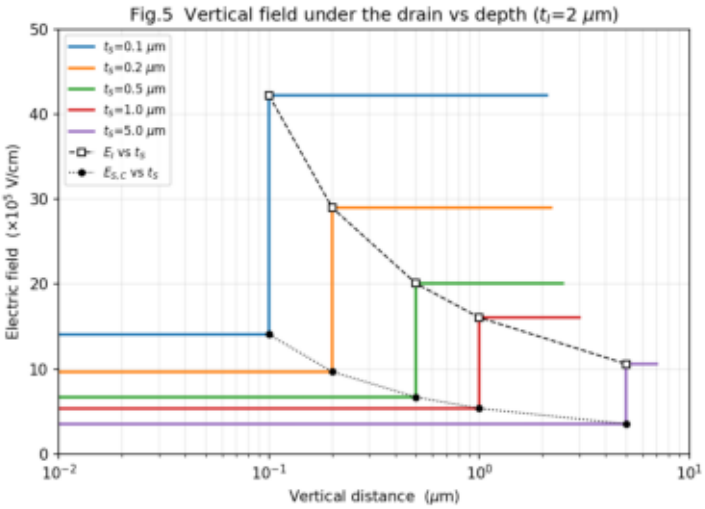
U-shaped family (t_I=2um); higher sigma_in / lower eps_I lift V_B,V. The conventional E_S,C=20 V/um curve (bottom) is far lower for thin silicon. Model 348/852 V at t_S=1/0.1um match the paper exactly.

Fig. 5 -- vertical field under the drain vs depth

Paper (original)



This reproduction



Each profile: silicon field limited to E_S,C, then jumps to the plateau E_I across the buried dielectric. The two envelopes (E_I and E_S,C vs t_S) rise sharply for thin silicon.

Notes, erratum and files

What matches

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The closed-form ENDIF quantities (E_S,C, alpha_S,C, E_I, V_B,V,
enhancement factors) reproduce Figs 2-5 and every stated number
to within rounding.
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Erratum found in the paper's text (p.2329)

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"t_S=1um, eps_I=3.9eps0: E_I = 161 and 681 V/um for sigma_in of
0 and 4e12" is inconsistent:
  * eq.(11) gives E_I = 161 + 186 = 347 V/um
  * the paper's own V_B,V=719 V implies the same 347
  * the paper's own Fig.3(a) sigma=4 curve reads ~350 V/um
Origin: using the LOW-K eps_I=2.0eps0 gives 319+362 = 681 exactly
-- the LK value was carried into the interface-charge example.
Text typo only; equations and figures are mutually consistent.
(See the slide deck zhang2009_erratum.pptx and verify_681.py.)
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Files

```
code/zhang2009_full.py        model -> Figs 2-5 + self-check
code/verify_681.py            three-way erratum verification
code/build_pdfpages_report.py this report
report/zhang2009_report.tex   LaTeX version (same content)
report/zhang2009_erratum.pptx annotated erratum slides
```